



Shore

**Year 12
Mathematics
Trial Examination
2010**

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black or blue pen
- Board-approved calculators may be used
- A table of standard integrals is provided on the back page of this question paper
- All necessary working should be shown in every question

Note: Any time you have remaining should be spent revising your answers.

Student Number:

Set:

Total marks – 120

- Attempt Questions 1 – 10
- All questions are of equal value
- Start each question in a new writing booklet
- Write your examination number on the front cover of each booklet to be handed in
- If you do not attempt a question, submit a blank booklet marked with your examination number and “N/A” on the front cover

Total Marks – 120
Attempt Questions 1–10
All questions are of equal value

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

Question 1 (12 marks)

Marks

- (a) Evaluate e^{-3} correct to 3 significant figures. 2
- (b) Factorise $8x^3 - 125$. 2
- (c) Simplify $\frac{5x-3}{x^2-9} - \frac{2}{x-3}$. 2
- (d) Find the values of x for which $|x+1| \leq 4$. 2
- (e) Find the integers a and b such that $(5 - \sqrt{2})^2 = a - b\sqrt{2}$. 2
- (f) Calculate the limiting sum of the geometric series $\frac{5}{6} + \frac{5}{36} + \frac{5}{216} + \dots$. 2

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Question 2 (12 marks) Use a SEPARATE writing booklet

Marks

(a) Differentiate with respect to x :

(i) $(x^3 + 7)^4$

2

(ii) $x \sin x$

2

(iii) $\frac{e^x}{2x+1}$

2

(b) Find $\int (\sec^2 3x + x) dx$.

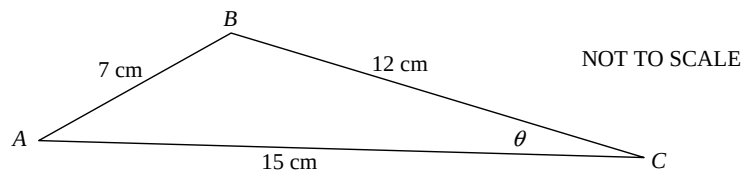
2

(c) Evaluate $\int_0^1 \frac{dx}{x+2}$.

2

(d)

2



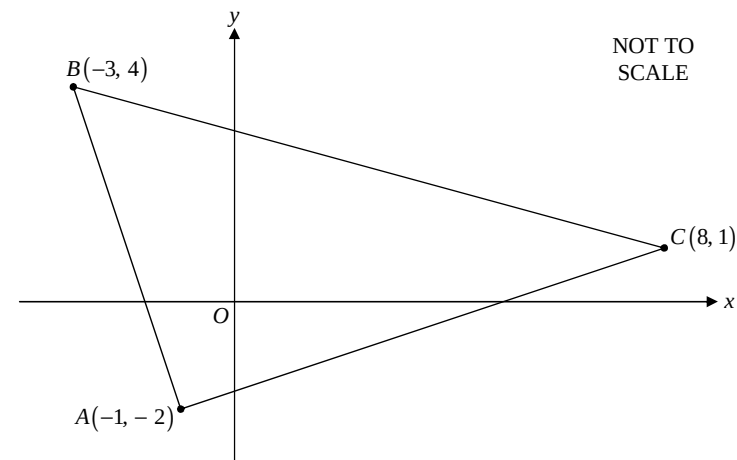
The diagram shows $\triangle ABC$ with $\angle ACB = \theta$, $AB = 7$ centimetres, $BC = 12$ centimetres and $AC = 15$ centimetres.

Find the value of θ correct to the nearest degree.

Question 3 (12 marks) Use a SEPARATE writing booklet

Marks

(a)



The diagram shows the points $A(-1, -2)$, $B(-3, 4)$ and $C(8, 1)$.

(i) Find the gradient of AB .

1

(ii) Show that AB is perpendicular to AC .

2

(iii) Find the length of the interval AC .

1

(iv) Hence, or otherwise, calculate the area of the triangle ABC .

2

(b) Find the equation of the tangent to the curve $y = 3e^{2x}$ at the point on the curve where $x = \frac{1}{2}$.

3

(c) Let α and β be the solutions of $x^2 - 3x + 7 = 0$.

(i) Find $\alpha\beta$.

1

(ii) Find $\alpha + \beta$.

1

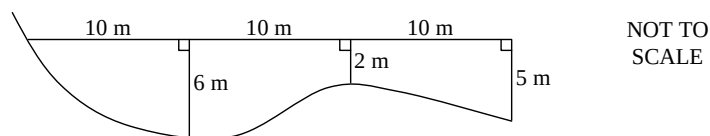
(iii) Hence, find $\frac{1}{\alpha} + \frac{1}{\beta}$.

1

Question 4 (12 marks) Use a SEPARATE writing booklet

Marks

- (a) Find the values of k for which the quadratic equation $5x^2 - 2x + k = 0$ has no real roots. 2
- (b) Four red marbles and five green marbles are contained in a cloth bag. Two marbles are randomly selected without replacement.
- (i) Find the probability of selecting two marbles of the same colour. 2
- (ii) Find the probability of selecting two marbles of different colours. 1
- (c) The diagram below shows the cross-section of a river with the depths of the water shown in metres, at 10 metre intervals.

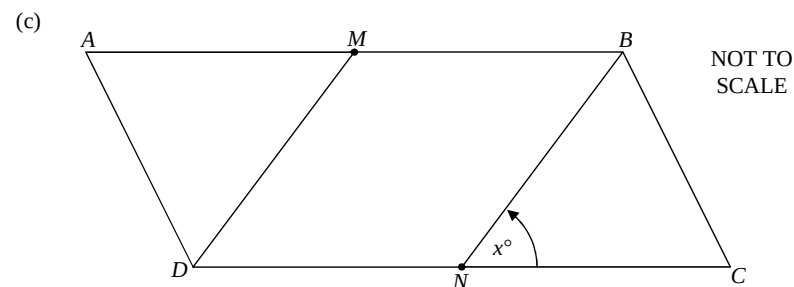


- (i) Use the trapezoidal rule to find an approximate value for the area of the cross-section. 2
- (ii) Water flows through this section of the river at a speed of 0.6 metres per second.
- Calculate the approximate volume of water that flows through this cross-section in one hour.
- (d) Consider the parabola $8y = x^2 - 6x - 23$.
- (i) Find the coordinates of the vertex. 2
- (ii) Find the coordinates of the focus. 1

Question 5 (12 marks) Use a SEPARATE writing booklet

Marks

- (a) State the domain of the function $y = \sqrt{36 - x^2}$. 1
- (b) Fred is training for a big running race. On the first day he runs 5 kilometres. On each subsequent day he runs 200 metres further than he did on the previous day. He stops training on the day he runs 42.2 kilometres.
- (i) How far does Fred run on the 50th day? 2
- (ii) How many days does Fred train for? 1
- (iii) What is the total distance that Fred runs during his training? 2



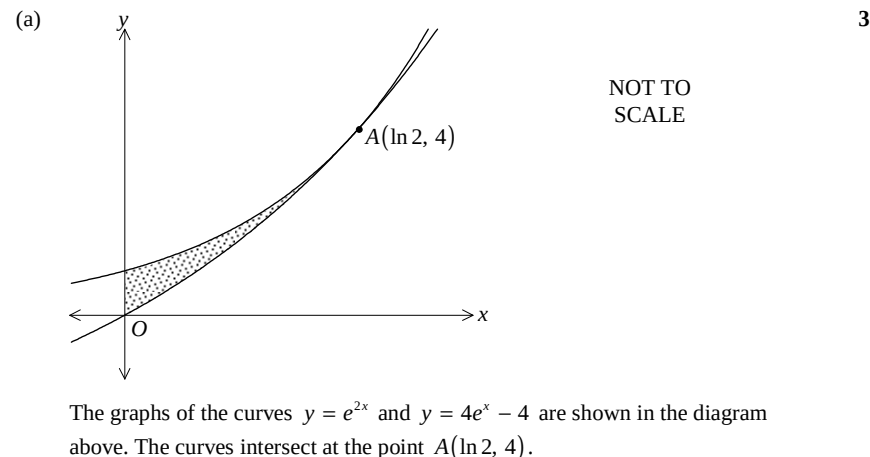
In the diagram, $DMBN$ is a rhombus. M and N are the midpoints of AB and CD respectively and $\angle CNB = x^\circ$.

Copy or trace the diagram into your writing booklet.

- (i) Show that $\angle AMD = x^\circ$, giving reasons. 2
- (ii) Prove that $\triangle AMD \equiv \triangle CNB$. 3
- (iii) Prove that $ABCD$ is a parallelogram. 1

Question 6 (12 marks) Use a SEPARATE writing booklet**Marks**

- (a) Solve $2\sin^2 x - 7\sin x + 3 = 0$ for $0 \leq x \leq 2\pi$. 3
- (b) (i) Find $\frac{d}{dx}[\log_e(\sin 2x)]$. 2
- (ii) Hence, or otherwise, evaluate $\int_{\frac{\pi}{8}}^{\frac{\pi}{4}} \cot 2x \, dx$. 2
- (c) Consider the function $f(x) = x^3 - 3x^2 + 8$.
- (i) Find the coordinates of the stationary points of the function $y = f(x)$, and determine their nature. 3
- (ii) Sketch the function $y = f(x)$, showing the stationary points. 1
- (iii) Find the values of x for which the curve $y = f(x)$ is concave down. 1

Question 7 (12 marks) Use a SEPARATE writing booklet**Marks**

Calculate the exact area of the shaded region.

- (b) A tank initially holds 2500 litres of water. The water drains from the bottom of the tank. The tank takes 50 minutes to empty.

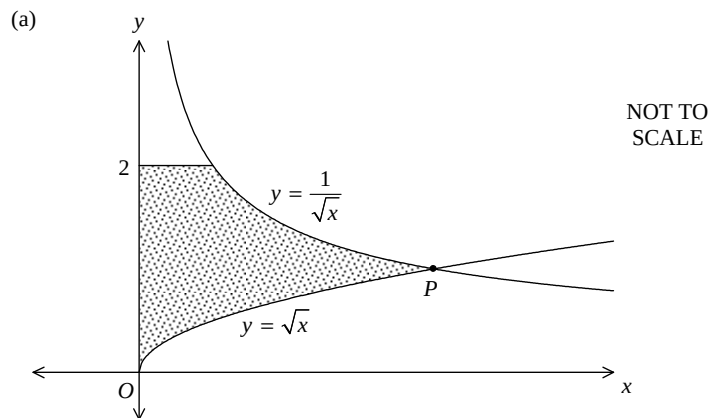
A mathematical model predicts that the volume, V litres, of water that will remain in the tank after t minutes is given by

$$V = 2500 \left(1 - \frac{t}{50} \right)^2, \text{ where } 0 \leq t \leq 50.$$

- (i) What volume does the model predict will remain after 10 minutes? 1
- (ii) At what rate does the model predict that the water will drain from the tank after 20 minutes? 2
- (iii) At what time does the model predict that the water will drain at its fastest rate from the tank? 2
- (c) A superannuation fund pays interest at the rate of 5% per annum compounding annually. Steven decides to invest \$7000 into the fund at the beginning of each year, commencing on the 1st of January 2011.
- (i) Write an expression for the value of Steven's fund after 3 years. 1
- (ii) What will be the value of Steven's superannuation when he retires on the 31st of December 2041? 3

Question 8 (12 marks) Use a SEPARATE writing booklet

Marks

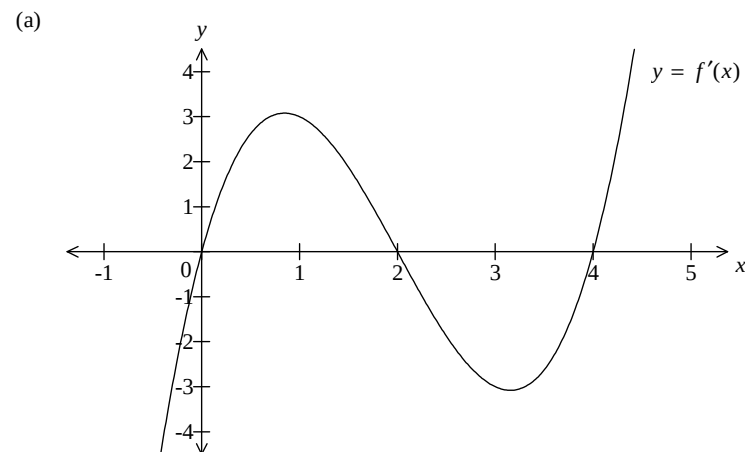


The graphs of the curves $y = \sqrt{x}$ and $y = \frac{1}{\sqrt{x}}$ intersect at the point P , as shown in the diagram above.

- (i) Show that P is the point $(1, 1)$. 1
- (ii) Find the area of the shaded region bounded by $y = \sqrt{x}$, $y = \frac{1}{\sqrt{x}}$, the y -axis and the line $y = 2$. 3
- (b) Solve $\log_7 x - \log_7 4 = 2 \log_7 3$. 2
- (c) The velocity of a particle is given by $v = 3 - 6 \cos t$ for $0 \leq t \leq 2\pi$, where v is measured in metres per second and t is the time in seconds.
- (i) Sketch the graph of v as a function of t for $0 \leq t \leq 2\pi$. 2
- (ii) At what times during this period is the particle at rest? 2
- (iii) Find an expression for the acceleration, $a \text{ m/s}^2$, in terms of t . 1
- (iv) Find when the particle first reaches its maximum acceleration. 1

Question 9 (12 marks) Use a SEPARATE writing booklet

Marks



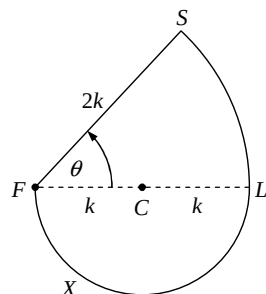
The above diagram shows a sketch of the gradient function of the curve $y = f(x)$.

In your writing booklet, draw a possible sketch of the function $y = f(x)$ given that $f(0) = 1$.

- (b) The radioisotope Technetium-99m is used for medical procedures and is produced at Lucas Heights in NSW. Technetium-99m has a rate of decay that is proportional to the mass M present at any given time, such that $\frac{dM}{dt} = -kM$.
- (i) Show that $M = M_0 e^{-kt}$, where k and M_0 are constants, satisfies the differential equation above. 1
- (ii) Technetium-99m has a half life of 6 hours. That is, the time taken for half the initial mass to decay is 6 hours. Find the value of k . 2
- (iii) A sample of Technetium-99m was shipped from the production site to a hospital in Western Australia. The total shipping time was 15.6 hours. 2
- How many kilograms were shipped if just **one** kilogram of Technetium-99m arrived at the hospital?

Question 9 continues

(c)



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A cam is formed with cross-section as shown in the diagram. The cross-section consists of a semi-circle FLX , with centre C and radius k , and a sector FSL , with centre F , radius $2k$ and angle θ radians.

- (i) Show that the perimeter P of the cam is given by $P = k(2\theta + \pi + 2)$. 1
- (ii) The area of the cross-section is 1 unit^2 . Find an expression for θ in terms of k . 2
- (iii) Hence, show that the perimeter P is given by 2

$$P = \frac{1}{k} + k\left(2 + \frac{\pi}{2}\right).$$

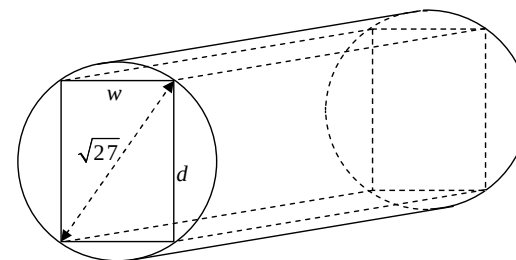
End of Question 9

Question 10 (12 marks) Use a SEPARATE writing booklet

Marks

- (a) A rectangular beam of width w cm and depth d cm can be cut from a cylindrical log of wood as shown in the diagram below.

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The diameter of the cross-section of the log (and hence the diagonal of the cross-section of the beam) is $\sqrt{27}$ cm.

The strength S of the beam is proportional to the product of its width and the square of its depth, so that $S = kd^2w$, where k is a positive constant.

- (i) Show that $S = k(27w - w^3)$. 2
- (ii) What numerical dimensions will give a beam of maximum strength? Leave your answer as an exact value. 3
- (iii) A square beam with diagonal of $\sqrt{27}$ cm is to be cut from an identical log. Show that the rectangular beam of maximum strength is more than 8% stronger than this square beam. 2
- (b) Consider the function $f(x) = x((\ln x)^2 - 2\ln x + 2)$.
- (i) Show that $f'(x) = (\ln x)^2$. 3
- (ii) Hence, or otherwise, find the volume of the solid of revolution formed when the region bounded by the curve $y = \ln x$ and the x -axis between $x = 1$ and $x = e$ is rotated about the x -axis. 2

END OF EXAM

Question 1

$$a) e^{-3} = 0.049787... \\ \approx 0.0498$$

$$b) 8x^3 - 125 = (2x)^3 - (5)^3 \\ = (2x - 5)(2x)^2 + 2x \times 5 + 5^2 \\ = (2x - 5)(4x^2 + 10x + 25)$$

$$c) \frac{5x-3}{(x^2-9)} - \frac{2}{x-3} = \frac{5x-3}{(x+3)(x-3)} - \frac{2(x+3)}{(x+3)(x-3)} \\ = \frac{5x-2x-3-6}{(x+3)(x-3)} \\ = \frac{3x-9}{(x+3)(x-3)} \\ = \frac{3(x-3)}{(x+3)(x-3)} \\ = \frac{3}{(x+3)}$$

$$d) |x+1| \leq 4 = -4 \leq (x+1) \leq 4 \\ -5 \leq x \leq 3$$

$$e) (5-\sqrt{2})^2 = (5-\sqrt{2})(5-\sqrt{2}) \quad \therefore a=27 \\ = 25 - 10\sqrt{2} + 2 \quad b=10 \\ = 27 - 10\sqrt{2}$$

$$f) a = \frac{5}{6} \quad r = \frac{1}{6} \quad S_{\infty} = \frac{a}{1-r} = \frac{\frac{5}{6}}{1-\frac{1}{6}} = \frac{\frac{5}{6}}{\frac{5}{6}} = 1$$

Question 2

$$a) i) \frac{d}{dx} (x^3+7)^4 = 4(x^3+7)^3 \times 3x^2 \\ = 12x^2(x^3+7)^3$$

$$ii) \frac{d}{dx} (x \sin x) = x \cos x + 1 \times \sin x \quad u=x \quad v=\sin x \\ = x \cos x + \sin x \quad u'=1 \quad v'=\cos x$$

$$iii) \frac{d}{dx} \left(\frac{e^x}{2x+1} \right) = \frac{(2x+1)e^x - 2e^x}{(2x+1)^2} \quad u=e^x \quad v=2x+1 \\ w=e^{2x} \quad v'=2 \\ = \frac{e^x(2x-1)}{(2x+1)^2}$$

$$b) \int (\sec^2 3x + x) dx = \frac{1}{3} \tan 3x + \frac{x^2}{2} + C$$

$$c) \int_0^1 \frac{dx}{x+2} = [\ln(x+2)]_0^1 \\ = \ln 3 - \ln 2 \\ = \ln \frac{3}{2}$$

$$d) \cos \theta = \frac{a^2 + b^2 - c^2}{2ab} \\ = \frac{12^2 + 15^2 - 7^2}{2 \times 12 \times 15} \\ = \frac{320}{360} \\ \theta = \cos^{-1} \left(\frac{320}{360} \right) \\ = 27.2660 \approx 27^\circ$$

Question 3

$$a) i) m_{AB} = \frac{4 - (-2)}{-3 - (-1)}$$

$$= \frac{6}{-2}$$

$$= -3$$

$$ii) m_{AC} = \frac{1 - (-2)}{8 - (-1)}$$

$$= \frac{3}{9}$$

$$= \frac{1}{3}$$

$$m_{AC} \times m_{AB} = -3 \times \frac{1}{3}$$

$$= -1$$

$$\therefore AC \perp AB$$

$$iii) d_{AC} = \sqrt{(8 - (-1))^2 + (1 - (-2))^2}$$

$$= \sqrt{9^2 + 3^2}$$

$$= \sqrt{81 + 9}$$

$$= \sqrt{90}$$

$$iv) d_{AB} = \sqrt{(-3 - (-1))^2 + (4 - (-2))^2}$$

$$= \sqrt{4 + 36}$$

$$= \sqrt{40}$$

$$\text{Area}_{\triangle ABC} = \frac{1}{2} \times \sqrt{90} \times \sqrt{40}$$

$$= 30 \text{ units}^2$$

Question 3 continued

$$b) y = 3e^{2x}$$

$$y' = 6e^{2x}$$

$$\text{when } x = \frac{1}{2}$$

$$y = 3e^{2 \times \frac{1}{2}}$$

$$= 3e$$

$$y' = 6e^{2 \times \frac{1}{2}}$$

$$= 6e$$

The eqn of a straight line is given by;

$$y - y_1 = m(x - x_1)$$

$$y - 3e = 6e(x - \frac{1}{2})$$

$$y - 3e = 6ex - 3e$$

$$y = 6ex$$

$$c) x^2 - 3x + 7 = 0$$

$$i) \alpha\beta = \frac{c}{a}$$

$$= \frac{7}{1}$$

$$= 7$$

$$ii) \alpha + \beta = \frac{-b}{a}$$

$$= \frac{-(-3)}{1}$$

$$= 3$$

$$iii) \frac{1}{\alpha} + \frac{1}{\beta} = \frac{\alpha + \beta}{\alpha\beta}$$

$$= \frac{3}{7}$$

Question 4

a) $5x^2 - 2x + k = 0$

For no real roots $\Delta < 0$

$$b^2 - 4ac < 0$$

$$(-2)^2 - 4 \times 5 \times k < 0$$

$$4 - 20k < 0$$

$$-20k < -4$$

$$k > \frac{1}{5}$$

b) i) $P(\text{same colour}) = P(RR) + P(GG)$
 $= \left(\frac{4}{9} \times \frac{3}{8}\right) + \left(\frac{5}{9} \times \frac{4}{8}\right)$

$$= \frac{12}{72} + \frac{20}{72}$$

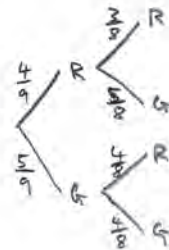
$$= \frac{32}{72}$$

$$= \frac{4}{9}$$

ii) $P(\text{different colour}) = 1 - P(\text{same colour})$

$$= 1 - \frac{4}{9}$$

$$= \frac{5}{9}$$



Question 4 continued

c) i) $\text{Area} = \frac{b}{2} [(h_1 + h_2) + 2(\text{middle})]$

$$= \frac{10}{2} [(0 + 5) + 2(6 + 2)]$$

$$= 5(5 + 16)$$

$$= 105 \text{ m}^2$$

ii) $\text{Volume} = 105 \text{ m}^2 \times 0.6 \text{ m s}^{-1} \times 3600$

$$= 226800 \text{ m}^3/\text{h}$$

$\therefore$ Approx 226800 m^3 flow through this cross-section in 1 hour.

d) i) $8y = x^2 - 6x - 23$

$$8y + 23 = x^2 - 6x$$

$$8y + 32 = x^2 - 6x + 9$$

$$8(y + 4) = (x - 3)^2$$

$$4a(y - k) = (x - h)^2$$

$\therefore$ Vertex at $(3, -4)$

ii) $4a = 8$

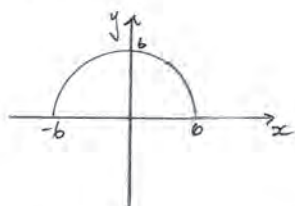
$$a = 2$$

$\therefore$ focal length = 2 and parabola is concave up

$\therefore$ focus at $(3, -2)$

Question 5

a) $y = \sqrt{36 - x^2}$



Domain is $-6 \leq x \leq 6$

b) $a = 5$ $d = 0.2$ $l = 42.2$

i) $T_n = a + (n-1)d$
 $T_{50} = 5 + (50-1)0.2$
 $T_{50} = 5 + 49 \times 0.2$
 $= 14.8 \text{ km}$

ii) $42.2 = 5 + (n-1)0.2$

$$\frac{37.2}{0.2} = n-1$$

$$n = \frac{37.2}{0.2} + 1$$

$$= 187$$

$\therefore$ Fred trains for 187 days.

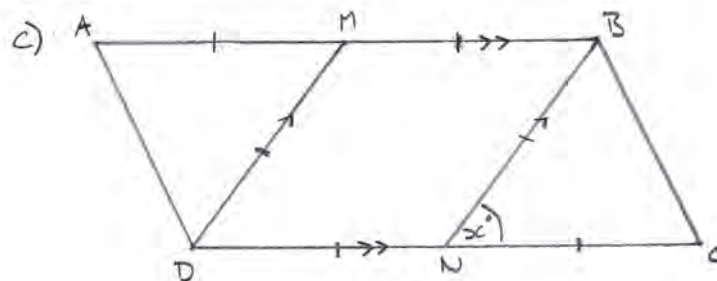
iii) $S_n = \frac{n}{2}(a+l)$

$$= \frac{187}{2}(5+42.2)$$

$$= 4413.2$$

$\therefore$ Fred runs a total distance of 4413.2 km.

Question 5 continued



i) $\angle MBN = x^\circ$ (alternate angles on // lines)

$\therefore \angle AND = x^\circ$ (corresponding angles on // lines)

ii) In $\triangle AMD$ and $\triangle CNB$

$DM = MB = BN = DN$ (sides of a rhombus)

$AM = MB$ (Given)

$NC = DN$ (Given)

$\therefore AM = NC$ (s)

$\angle AMD = \angle CNB$ (Proved in part (i)) (A)

$DM = BN$ (sides of a rhombus) (s)

$\therefore \triangle AMD \equiv \triangle CNB$

iii) $AB = 2AM$ (M is midpoint of AB)

$CD = 2NC$ (N is midpoint of CD)

Since $AM = NC$ (proved in part (ii))

$\therefore AB = CD$

$AD = CB$ (matching sides of congruent triangles)

$\therefore$ ABCD is a parallelogram
 (two pairs of equal opposite sides)

Question 6

a) $2 \sin^2 x - 7 \sin x + 3 = 0$ let $u = \sin x$

$$2u^2 - 7u + 3 = 0$$

$$(2u-1)(u-3) = 0$$

$$\therefore u = 3 \quad \text{or} \quad u = \frac{1}{2}$$

$$\therefore \sin x = 3 \quad \text{or} \quad \sin x = \frac{1}{2}$$

No soln

as $|\sin x| \leq 1$
for all x

$$\therefore x = \frac{\pi}{6} \quad \text{or} \quad \pi - \frac{\pi}{6}$$

$$= \frac{\pi}{6} \quad \text{or} \quad \frac{5\pi}{6}$$

b) i) $\frac{d}{dx} [\log_e (\sin 2x)] = \frac{2 \cos 2x}{\sin 2x}$

$$= 2 \cot 2x$$

ii) $\int_{\pi/8}^{\pi/4} \cot 2x \, dx = \frac{1}{2} \int_{\pi/8}^{\pi/4} 2 \cot 2x \, dx$

$$= \frac{1}{2} [\log_e (\sin 2x)]_{\pi/8}^{\pi/4}$$

$$= \frac{1}{2} [\log_e (\sin 2 \cdot \frac{\pi}{4}) - \log_e (\sin 2 \cdot \frac{\pi}{8})]$$

$$= \frac{1}{2} [\ln 1 - \ln \frac{1}{\sqrt{2}}]$$

$$= \frac{1}{2} [0 - (\ln 1 - \ln \sqrt{2})]$$

$$= \frac{1}{2} (0 - 0 + \ln \sqrt{2})$$

$$= \frac{1}{2} \ln \sqrt{2}$$

$$= \frac{1}{4} \ln 2$$

Question 6 continued

c) $f(x) = x^3 - 3x^2 + 8$

i) $f'(x) = 3x^2 - 6x$

$$= 3x(x-2)$$

Stationary points when $f'(x) = 0$

$$\therefore 0 = 3x(x-2)$$

$$x = 0 \quad \text{and} \quad x = 2$$

$$f(0) = 0^3 - 3(0)^2 + 8 = 8 \quad f(2) = 2^3 - 3 \cdot 2^2 + 8 = 8 - 12 + 8 = 4$$

Stat pts at $(0, 8)$ and $(2, 4)$

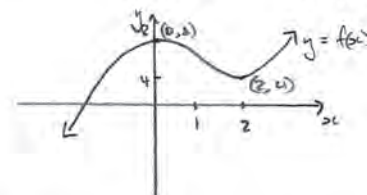
$$f''(x) = 6x - 6$$

$$f''(0) = 6 \cdot 0 - 6$$

$= -6 < 0$ $\therefore$ concave down at $(0, 8)$
 $\therefore (0, 8)$ is a maximum.

$f''(2) = 6 \cdot 2 - 6$ $\therefore$ concave up at $(2, 4)$
 $= 6 > 0$ $\therefore (2, 4)$ is a minimum.

ii)



iii) Concave down when $f''(x) < 0$
 $6x - 6 < 0 \quad \therefore f(x)$ is concave down
 $x < 1$ for $x < 1$

Question 7

$$\begin{aligned} \text{a) Area} &= \int_0^{\ln 2} e^{2x} - 4e^x + 4 \, dx \\ &= \left[\frac{e^{2x}}{2} - 4e^x + 4x \right]_0^{\ln 2} \\ &= \left(\frac{e^{2\ln 2}}{2} - 4e^{\ln 2} + 4\ln 2 \right) - \left(\frac{e^{2 \cdot 0}}{2} - 4e^0 + 0 \right) \\ &= 2 - 8 + 4\ln 2 - \frac{1}{2} + 4 + 0 \\ &= 4\ln 2 - \frac{5}{2} \end{aligned}$$

$$\text{b) i) } V = 2500 \left(1 - \frac{t}{50} \right)^2$$

$$\text{for } t = 10$$

$$V = 2500 \left(1 - \frac{10}{50} \right)^2$$

$$= 2500 \times \frac{16}{25}$$

$$= 1600 \text{ L}$$

$$\text{ii) } \frac{dV}{dt} = 2 \times 2500 \left(1 - \frac{t}{50} \right) \times -\frac{1}{50}$$

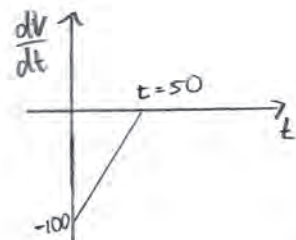
$$= -100 \left(1 - \frac{t}{50} \right)$$

$$= -100 + 2t$$

$$\begin{aligned} \text{when } t = 20 \quad \frac{dV}{dt} &= -100 + 2 \times 20 \\ &= -60 \text{ L/min} \end{aligned}$$

Question 7 continued

b) iii) $\frac{dv}{dt} = -100 + 2t$



the fastest rate at which water leaves the tank is at $t=0$ i.e. $\frac{dv}{dt} = -100 \text{ L/min}$

c) 31 DEC 2011 ($A_1 = 7000 \times 1.05$)

31 DEC 2012 ($A_2 = 7000 \times 1.05^2 + 7000 \times 1.05$)

31 DEC 2013 ($A_3 = 7000 \times 1.05^3 + 7000 \times 1.05^2 + 7000 \times 1.05$)

i) $A_3 = 7000 (1.05 + 1.05^2 + 1.05^3)$

ii) Assume $A_n = 7000 (1.05 + 1.05^2 + \dots + 1.05^n)$

i.e. Geometric Series $a = 1.05$
 $r = 1.05$
 $n = 31$

$$S_n = \frac{a(r^n - 1)}{r - 1}$$

$$S_{31} = \frac{1.05 \times (1.05^{31} - 1)}{0.05}$$

$$= 74.2988 \dots$$

$$\therefore A_{31} = \$7000 \times S_{31}$$

$$= \$520\,091.81$$

Question 8

a) i) $\sqrt{x} = \frac{1}{\sqrt{x}} \quad \therefore P$ is the point $(1, 1)$

$$1 = x^2$$

$$\therefore x = 1$$

$$y = \sqrt{1}$$

$$= 1$$

ii) Area = $\int_0^1 y^2 dy + \int_1^2 y^{-2} dy$

$$= \left[\frac{y^3}{3} \right]_0^1 + \left[\frac{y^{-1}}{-1} \right]_1^2$$

$$= \left(\frac{1}{3} - 0 \right) + \left(-\frac{1}{2} + 1 \right)$$

$$= \frac{5}{6} \text{ units}^2$$

b) $\log_7 x - \log_7 4 = 2 \log_7 3$

$$\log_7 \left(\frac{x}{4} \right) = \log_7 9$$

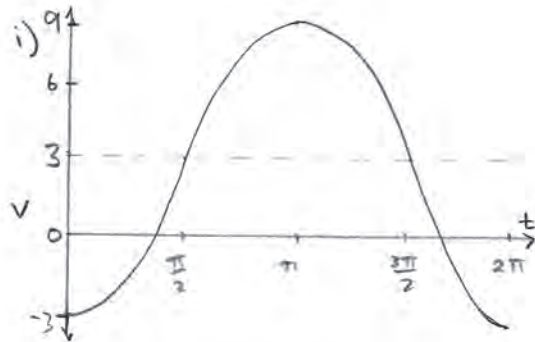
$$\frac{x}{4} = 9$$

$$x = 36$$

note: $y = \sqrt{x}$
 $x = y^2$
 $y = \frac{1}{\sqrt{x}}$
 $x = y^{-2}$

Question 8 continued

c) $v = 3 - 6 \cos t$



ii) Particle is at rest when $v = 0$

$$0 = 3 - 6 \cos t$$

$$6 \cos t = 3$$

$$\cos t = \frac{1}{2}$$



$$\frac{s}{r} = \frac{A}{C}$$

$$\therefore t = \frac{\pi}{3} \text{ or } 2\pi - \frac{\pi}{3}$$

$$= \frac{\pi}{3} \text{ or } \frac{5\pi}{3}$$

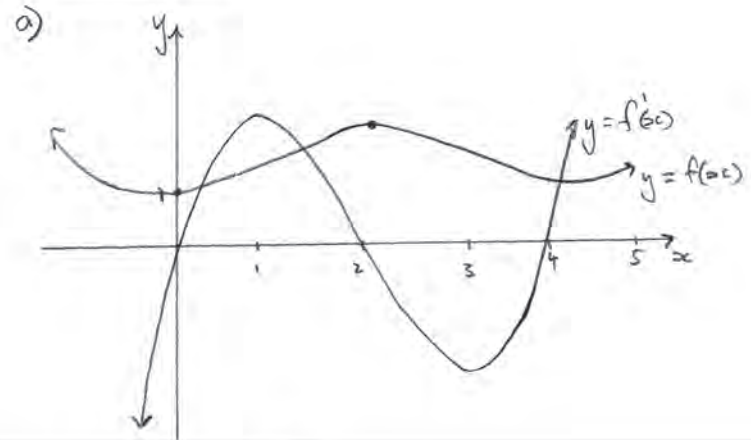
iii) $a = \frac{dv}{dt}$

$$\therefore a = 6 \sin t$$

iv) $\sin t$ has its first maximum when $t = \frac{\pi}{2}$

$\therefore$ The particle first reaches its maximum acceleration when $t = \frac{\pi}{2}$ seconds.

Question 9



b) i) $M = M_0 e^{-kt}$

$$\frac{dM}{dt} = M_0 \times -k \times e^{-kt}$$

$$= -k \times M_0 e^{-kt}$$

$$= -kM$$

ii) when $t = 6$ $\frac{M}{M_0} = \frac{1}{2}$

$$\frac{1}{2} = e^{-k \times 6}$$

$$\ln \frac{1}{2} = -6k$$

$$k = \frac{\ln 2}{6}$$

ii) $M = 1$, $k = \frac{\ln 2}{6}$, $t = 15.6$

$$1 = M_0 \times e^{-\frac{\ln 2}{6} \times 15.6}$$

$$= 6.06286...$$

$\therefore$ Approx 6kg was shipped

Question 9 continued

$$\begin{aligned} \text{c) i) } P &= 2k + 2k\theta + \pi k \\ &= k(2\theta + \pi + 2) \end{aligned}$$

ii) Area = Area of sector + Area of semicircle

$$= \frac{1}{2} r^2 \theta + \frac{1}{2} \pi R^2 \quad r = 2k$$

$$R = k$$

$$= \frac{1}{2} (2k)^2 \theta + \frac{1}{2} \pi k^2$$

$$1 = 2k^2 \theta + \frac{1}{2} \pi k^2$$

$$2k^2 \theta = 1 - \frac{1}{2} \pi k^2$$

$$\theta = \frac{1 - \frac{1}{2} \pi k^2}{2k^2}$$

$$= \frac{1}{2k^2} - \frac{\pi}{4}$$

$$\text{iii) for } \theta = \frac{1}{2k^2} - \frac{\pi}{4}$$

$$P = k(2\theta + \pi + 2)$$

$$= k\left(2\left(\frac{1}{2k^2} - \frac{\pi}{4}\right) + \pi + 2\right)$$

$$= k\left(\frac{1}{k^2} - \frac{\pi}{2} + \pi + 2\right)$$

$$= \frac{1}{k} + \frac{k\pi}{2} + 2k$$

$$= \frac{1}{k} + k\left(2 + \frac{\pi}{2}\right)$$

Question 10

$$\text{a) i) } S = k d^2 w$$

$$d \sqrt{27-w^2}$$

$$\therefore S = k(27 - w^2)w$$

$$= k(27w - w^3)$$

$$d^2 + w^2 = 27$$

$$d^2 = 27 - w^2$$

$$\text{ii) } \frac{dS}{dw} = k(27 - 3w^2)$$

$$\frac{dS}{dw} = 0 \quad \text{for maximum strength}$$

$$0 = k(27 - 3w^2)$$

$$w^2 = 9$$

$$w = \pm 3$$

$$= 3 \quad (\text{ignore -ve dimension})$$

$$\text{when } w = 3, \quad d^2 = 27 - 9$$

$$= 18$$

$$d = \sqrt{18}$$

$$\text{check max } \frac{d^2 S}{dw^2} = -6wk < 0 \quad (\text{assume } w > 0)$$

($k > 0$ given)

$$\therefore \text{max strength when } w = 3 \text{ cm} + d = \sqrt{18} \text{ cm}$$

$$\text{iii) } 2w^2 = 27 \quad \therefore d = w = \sqrt{\frac{27}{2}}$$

$$w^2 = \frac{27}{2}$$

Question 10 continued

$$\begin{aligned} \text{a) iii) } S_{\text{square}} &= kd^2 \omega \\ &= k \times \frac{27}{2} \times \frac{\sqrt{27}}{2} \end{aligned}$$

$$\begin{aligned} S_{\text{max}} &= kd^2 \omega \\ &= k \times 18 \times 3 \end{aligned}$$

$$\begin{aligned} \frac{S_{\text{max}}}{S_{\text{square}}} &= \frac{54k}{\frac{27k}{2} \times \frac{\sqrt{27}}{2}} \\ &= \frac{4}{\sqrt{\frac{27}{2}}} \\ &= 1.08866... \end{aligned}$$

$S_{\text{max}} > 1.08 \times S_{\text{square}}$ as Required.

$$\text{b) i) } f(x) = x((\ln x)^2 - 2\ln x + 2)$$

$$f'(x) = x\left(\frac{2\ln x}{x} - \frac{2}{x}\right) + ((\ln x)^2 - 2\ln x + 2)$$

$$= 2\ln x - 2 + (\ln x)^2 - 2\ln x + 2$$

$$= (\ln x)^2$$

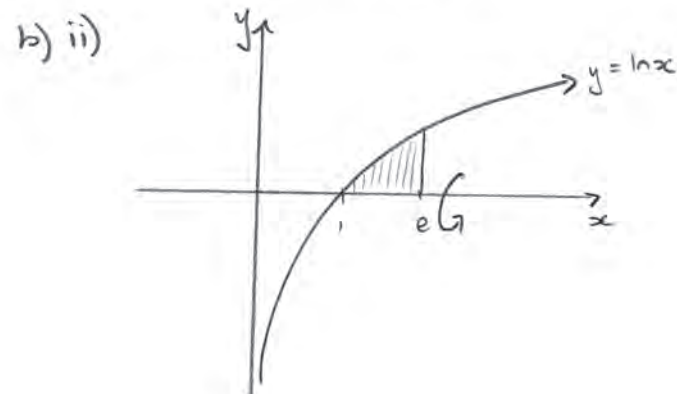
$$u = x$$

$$u' = 1$$

$$v = (\ln x)^2 - 2\ln x + 2$$

$$v' = \frac{2\ln x}{x} - \frac{2}{x}$$

Question 10 continued



$$V = \pi \int_1^e y^2 dx$$

$$= \pi \int_1^e (\ln x)^2 dx$$

$$= \pi \left[x((\ln x)^2 - 2\ln x + 2) \right]_1^e$$

$$= \pi [e(1^2 - 2 + 2) - 1(0^2 - 0 + 2)]$$

$$= \pi (e - 2) \text{ units}^3$$